

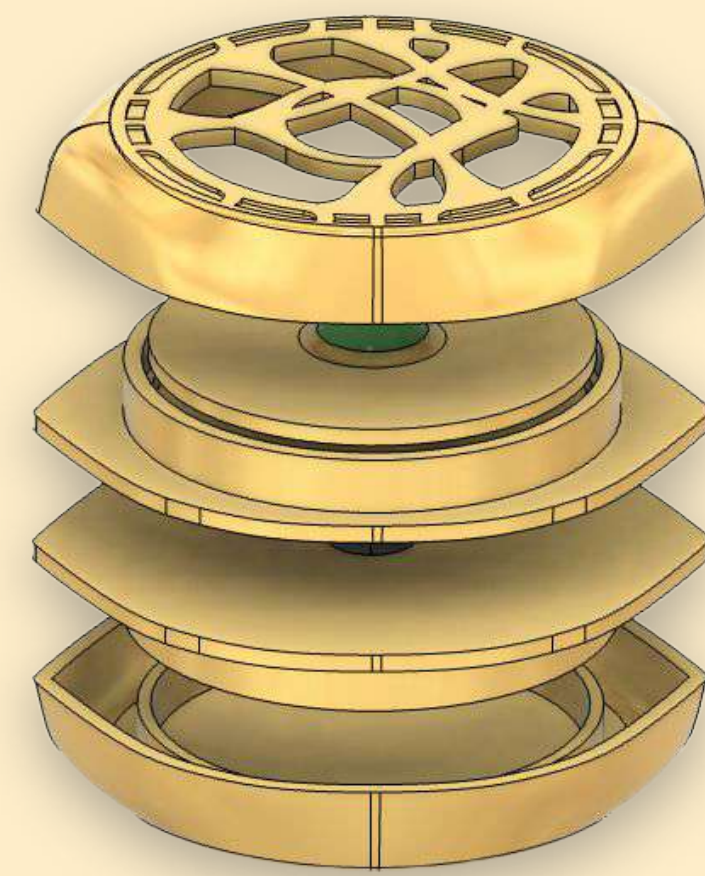


THE EYE OF AGAMOTTO

MASTERS OF THE MYSTIC YO-YOS (TEAM I): ADELENE CHAN, ELIJAH YOUNG, HECTOR LUGARO, KENNETH ORANGA, NICOLE LIN

JOURNEY

Inspiration: The Eye of Agamotto, a magical amulet used by Doctor Strange



- Design Goals:**
- Oblong, eye shape with smooth edges
 - Fine engravings and overhanging branches
 - Green thermoformed "Time Stone"

DESIGN FOR MANUFACTURING



9/17: Initial

10/12

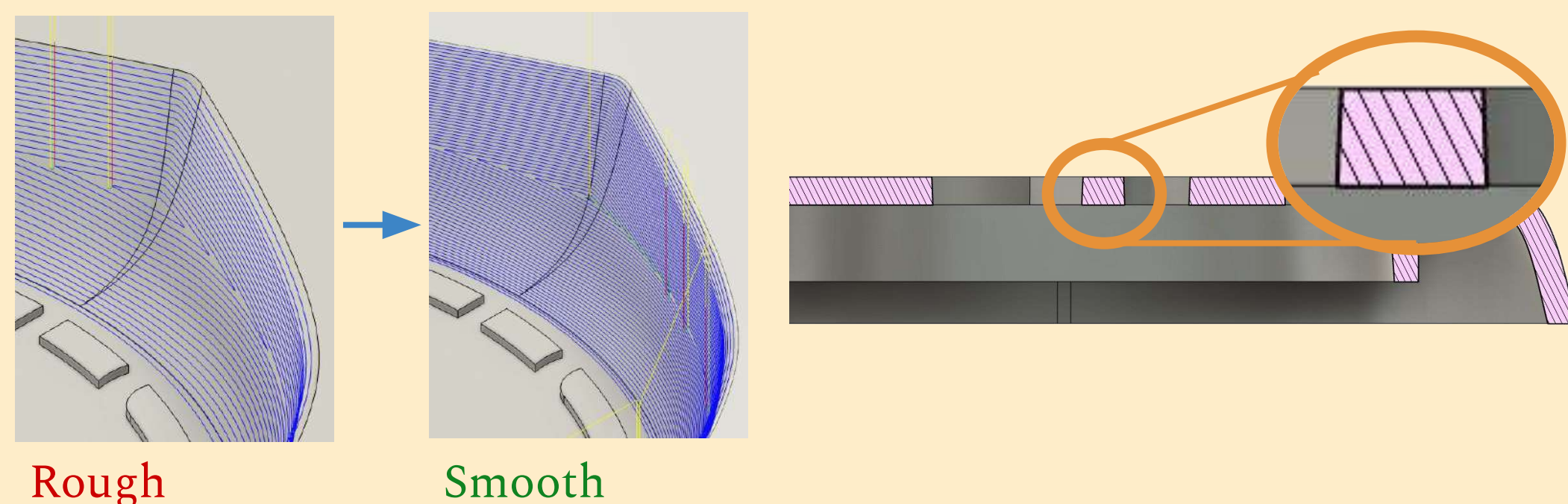
10/21: Final

Challenge #1

- Fine details limited by end mill diameter (1/16") & need surface area for ejector pins → widen channels

Challenge #2

- Edges/many surfaces may lead to part getting stuck in mold → smoother surface finish, draft angles of 2°



Rough

Smooth

MANUFACTURING

Injection Molding: used the Arburg Injection Molder to produce 50 top rings, 50 bottom rings, and 100 body parts using 3 aluminum molds



Thermoform: used thermoformer, punch, & spray paint to produce 50 parts with SLA printed mold



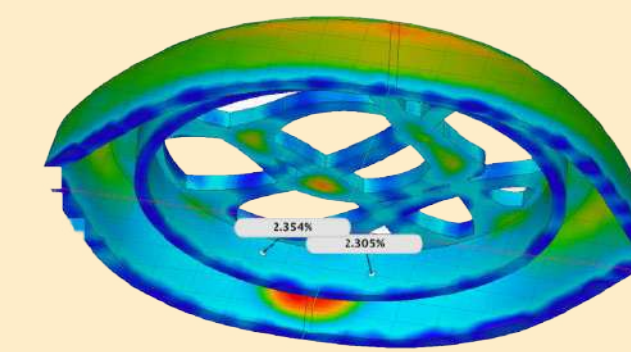
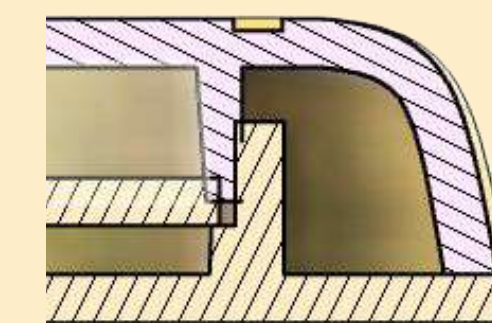
DIFFICULTIES



- **Most Difficult:** Machining detailed ring molds
 - Tool breakage, rough surface → decreased material removal rate
 - Unable to machine narrow groove → widen channels
- **Additional Difficulties**
 - **Bottom Ring:** Oversized pressfit → scale for shrinkage
 - **Body:** IM burning → lowered velocity to 0.5 m/s and pressure to 400 psi
 - **Thermoform Part:** Difficult to center for die-cutter → sharp "cliff" edge in mold for alignment

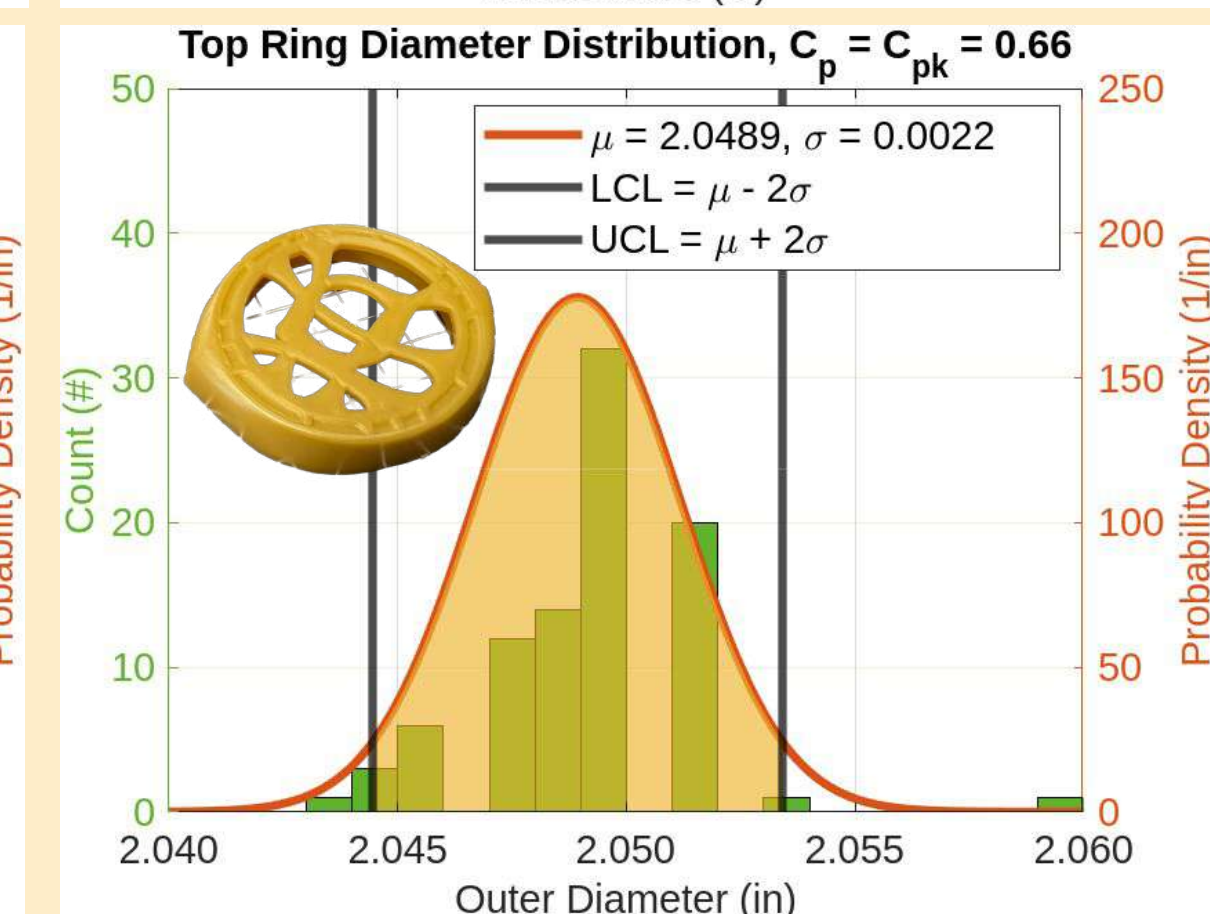
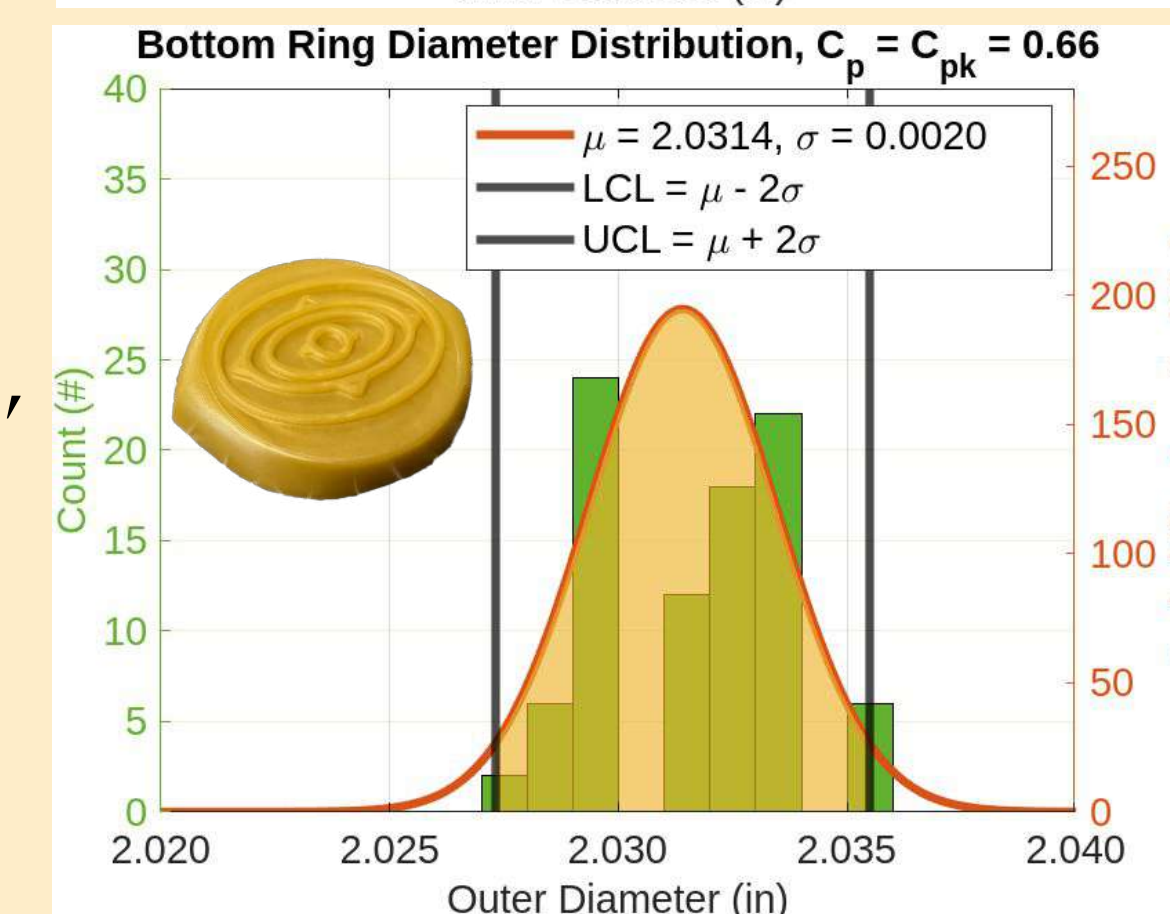
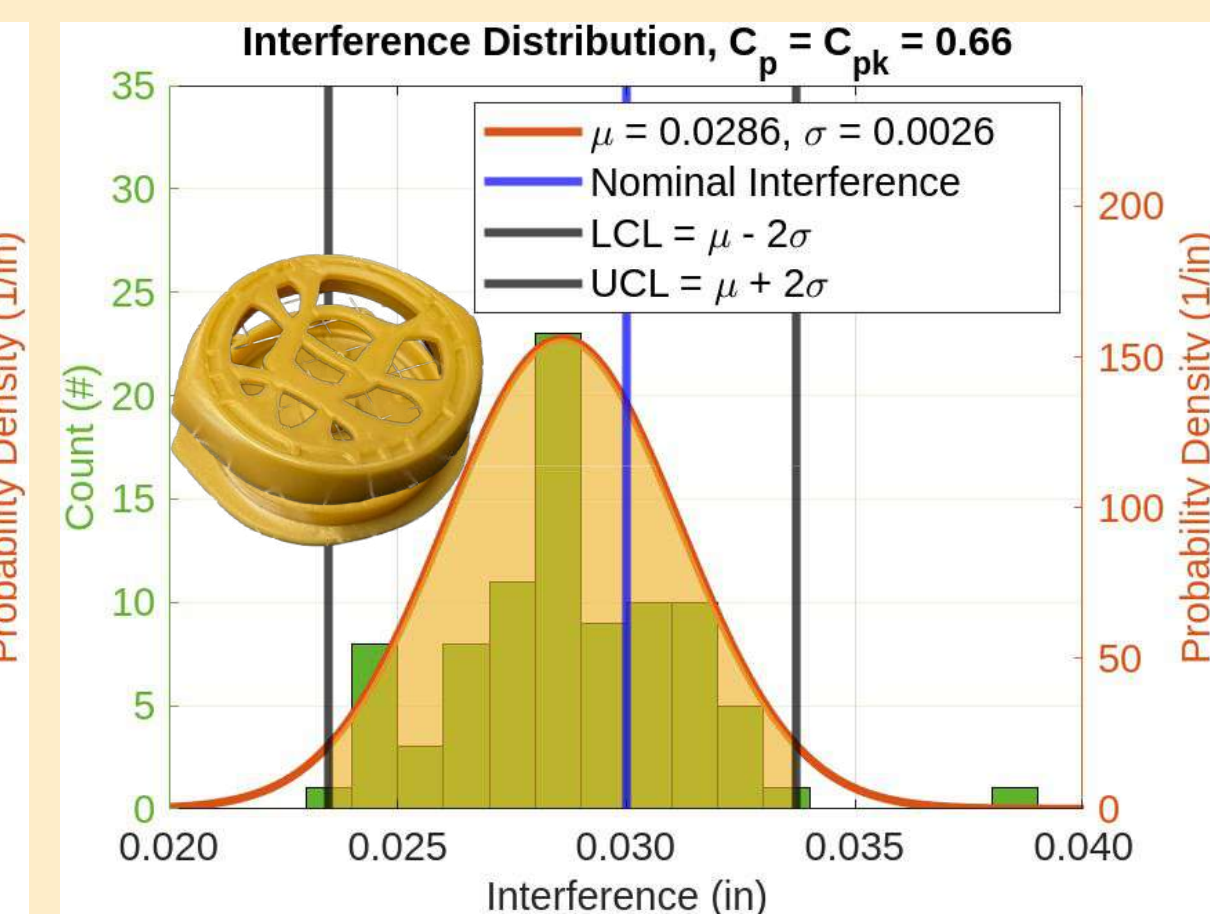
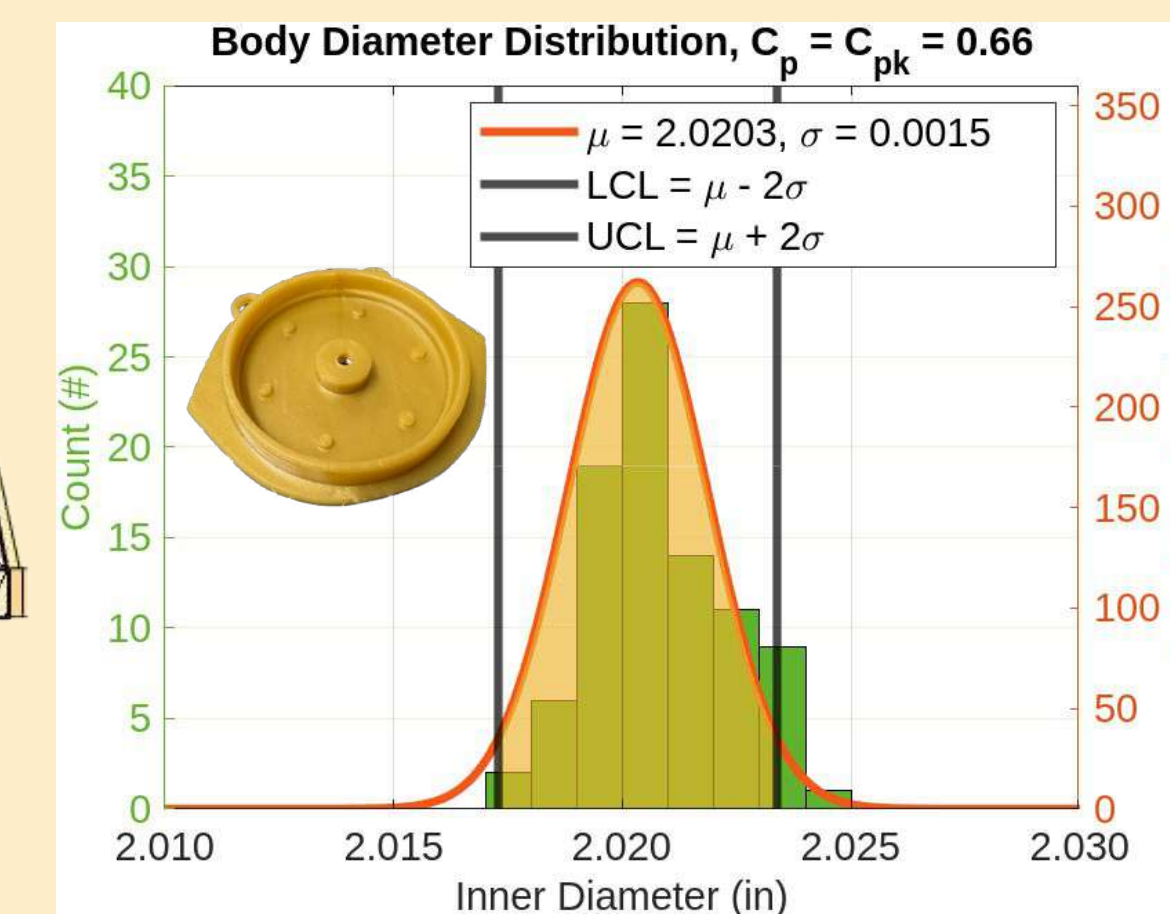
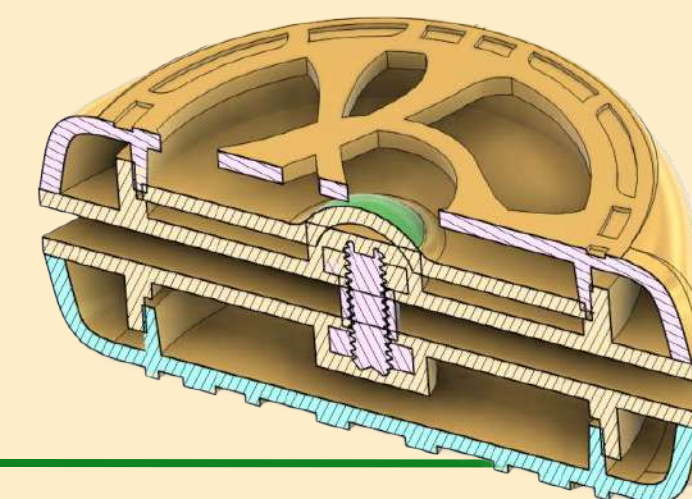
PRESS FIT ANALYSIS

Press Fit Design: Used a commonly accepted value of 0.015" and ensured correct machined dimensions via shrinkage simulations and analysis



Press Fit Outcomes: Our measurements differed from our intended specification limits (SL). However, since both parts had the same shrinkage, the mean interference was close to nominal.

- **Body SL:** [2.02, 2.04]
- **Ring SL:** [1.99, 2.01]



LESSONS LEARNED

- Account for machine downtime in manufacturing and know your cycle times
- Iteration is the learning process
- Consider manufacturability and ease of assembly from the start
- Thorough documentation helps everyone on the team